

Paper 1 - Set 09 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	D	Quadratic functions
2	F	Polynomials, factor and remainder theorems
3	B	Trigonometric functions and equations
4	B	Angles, polygons, congruence and vectors
5	A	Functions and transformations
6	B	Ratio, proportion and percentages
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10	C	Integration
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Worked solutions

Question 1 QUA-13 · Quadratic functions · spec MM1.3

Which of the following quadratics has its vertex in the second quadrant (where $x < 0$ and $y > 0$)?

Answer: D

Complete the square: A vertex $(-2, -3)$; B $(2, 1)$; C $(-1, -2)$; D $(-3, 1)$, which has $x < 0$ and $y > 0$; E $(1, 2)$. So D.

Full method - how and why

A 'which one' question about vertices. Complete the square on each option - but use the sign of the x -coordinate to eliminate most of them first.

- 1 Recall that for $y = x^2 + bx + c$ the vertex is at $x = -\frac{b}{2}$.
why: You only need the vertex position, not the whole completed square, to test the first condition $x < 0$.
- 2 Second quadrant needs $x < 0$, so $-\frac{b}{2} < 0$, i.e. $b > 0$: only $y = x^2 + 4x + 1$ and $y = x^2 + 6x + 10$ survive among the upward parabolas.
why: Eliminating on one condition first halves the work.
- 3 Compute the y -coordinates: $x^2 + 4x + 1 = (x + 2)^2 - 3$, vertex $(-2, -3)$: fails $y > 0$. $x^2 + 6x + 10 = (x + 3)^2 + 1$, vertex $(-3, 1)$: passes.
why: Now apply the second condition to the short list.
- 4 Check the downward ones quickly: $-x^2 - 2x - 3$ has vertex $(-1, -2)$ and $-x^2 + 2x + 1$ has vertex $(1, 2)$ - neither is in the second quadrant.
why: Worth 10 seconds of confirmation on a 'which one' question.

Where it goes wrong

- Reading the vertex of $(x + 3)^2 + 1$ as $(3, 1)$ - the sign flips.
- Mixing up the quadrants: the second quadrant is up and to the left.

Takeaway On 'which of these' questions, test the cheapest condition first and eliminate before doing any heavy algebra.

Question 2 POL-04 · Polynomials, factor and remainder theorems · spec MM1.6

Find the remainder when $x^4 + 2x^3 - 3x + 1$ is divided by $(2x - 1)$.

Answer: F

$$f\left(\frac{1}{2}\right) = \frac{1}{16} + \frac{2}{8} - \frac{3}{2} + 1 = \frac{1}{16} + \frac{4}{16} - \frac{24}{16} + \frac{16}{16} = -\frac{3}{16}.$$

Full method - how and why

A divisor of the form $ax - b$. The Remainder Theorem still applies, but the substitution value is $\frac{b}{a}$, not b .

- 1 $2x - 1 = 0$ at $x = \frac{1}{2}$.
why: The remainder on dividing by $(ax - b)$ is $f(b/a)$ - the root of the divisor, whatever the leading coefficient.
- 2 Substitute: $\left(\frac{1}{2}\right)^4 + 2\left(\frac{1}{2}\right)^3 - 3\left(\frac{1}{2}\right) + 1 = \frac{1}{16} + \frac{2}{8} - \frac{3}{2} + 1$.
why: Work each power out separately: $\frac{1}{16}$, $\frac{1}{8}$, then the linear and constant terms.
- 3 Put everything over 16: $\frac{1}{16} + \frac{4}{16} - \frac{24}{16} + \frac{16}{16} = -\frac{3}{16}$.
why: A common denominator prevents fraction slips; $\frac{2}{8} = \frac{4}{16}$ and $\frac{3}{2} = \frac{24}{16}$.

Where it goes wrong

- Substituting $x = 1$ or $x = 2$ instead of $\frac{1}{2}$.
- Arithmetic sign slips giving $+\frac{3}{16}$, which is offered.

Takeaway For divisor $ax - b$, evaluate at $x = \frac{b}{a}$; convert everything to a common denominator before adding.

Question 3 TRG-08 · Trigonometric functions and equations · spec MM4.6

Find the smallest positive solution of $\cos\left(x - \frac{\pi}{6}\right) = -\frac{1}{2}$.

Answer: B

$x - \frac{\pi}{6} = \frac{2\pi}{3}$ gives $x = \frac{5\pi}{6}$ (the other solution $x - \frac{\pi}{6} = \frac{4\pi}{3}$ gives $\frac{3\pi}{2}$, larger).

Full method - how and why

An equation with a shifted argument. Solve for the bracket first, then unshift - and be careful about which solution is smallest.

- 1 Let $u = x - \frac{\pi}{6}$ and solve $\cos u = -\frac{1}{2}$.
why: Substituting keeps the shift out of the way until the end.
- 2 $\cos u = -\frac{1}{2}$ at $u = \frac{2\pi}{3}$ and $u = \frac{4\pi}{3}$ (second and third quadrants).
why: $\cos$ is negative in quadrants 2 and 3; the reference angle is $\frac{\pi}{3}$.
- 3 Convert back: $x = u + \frac{\pi}{6}$, giving $x = \frac{2\pi}{3} + \frac{\pi}{6} = \frac{5\pi}{6}$ and $x = \frac{4\pi}{3} + \frac{\pi}{6} = \frac{3\pi}{2}$.
why: Common denominator 6: $\frac{4\pi}{3} + \frac{\pi}{6} = \frac{5\pi}{6}$.
- 4 The smallest positive one is $\frac{5\pi}{6}$.
why: Compare the candidates; $\frac{5\pi}{6} < \frac{3\pi}{2}$.

Where it goes wrong

- Subtracting the shift instead of adding when converting back.
- Using the reference angle $\frac{\pi}{3}$ directly, which gives $\frac{\pi}{2}$ - an offered option.

Takeaway Solve for the whole bracket, then undo the shift; adjust the interval for the bracket if the question restricts x .

Question 4 GEO-12 · Angles, polygons, congruence and vectors · spec M5.6

A shape is reflected in the line $y = x$ and then rotated 90° anticlockwise about the origin. The point $(3, 1)$ ends up at

Answer: B

Reflection in $y = x$: $(3, 1) \rightarrow (1, 3)$. Rotation 90° anticlockwise: $(x, y) \rightarrow (-y, x)$, so $(1, 3) \rightarrow (-3, 1)$.

Full method - how and why

Two transformations in sequence. Apply each rule in the stated order to the coordinates.

- 1 Reflection in $y = x$ swaps the coordinates: $(3, 1) \rightarrow (1, 3)$.
why: The line $y = x$ exchanges the roles of the two axes.
- 2 Rotation 90° anticlockwise about the origin sends $(x, y) \rightarrow (-y, x)$.
why: Worth memorising; check it on $(1, 0) \rightarrow (0, 1)$ ✓.
- 3 Apply it: $(1, 3) \rightarrow (-3, 1)$.
why: Here $x = 1$, $y = 3$, so the image is $(-3, 1)$.

Where it goes wrong

- Applying the transformations in the wrong order.
- Using the clockwise rule $(x, y) \rightarrow (y, -x)$, giving $(3, -1)$ - an offered option.

Takeaway Anticlockwise 90° : $(x, y) \rightarrow (-y, x)$. Clockwise: $(x, y) \rightarrow (y, -x)$. Order matters.

Question 5 FUN-06 · Functions and transformations · spec MM8.2

The curve $y = x^2 - 6x + 5$ is reflected in the y -axis. Find the equation of the reflected curve.

Answer: A

Reflecting in the y -axis replaces x with $-x$: $y = x^2 + 6x + 5$.

Full method - how and why

Reflection in the y -axis replaces x by $-x$. Only the odd powers change sign.

- 1 Replace x with $-x$: $y = (-x)^2 - 6(-x) + 5$.

why: Reflecting in the y -axis maps the point (x, y) to $(-x, y)$, which is achieved by substituting $-x$.

- 2 Simplify: $(-x)^2 = x^2$ and $-6(-x) = +6x$, so $y = x^2 + 6x + 5$.

why: Even powers are unaffected; odd powers flip sign. The constant is also unchanged.

Where it goes wrong

- Changing the sign of the constant too, giving $x^2 + 6x - 5$.
- Confusing it with reflection in the x -axis, which would negate the whole right-hand side.

Takeaway y -axis reflection: $x \rightarrow -x$, so only odd-degree terms change sign. x -axis reflection: $y \rightarrow -y$, so everything does.

Question 6 RAT-03 · Ratio, proportion and percentages · spec M3.4

£600 is divided between A , B and C in the ratio $2 : 3 : 7$. How much more does C receive than A ?

Answer: B

Each part is £50; $C - A = 5$ parts = £250.

Full method - how and why

Sharing in a ratio: find the value of one part, then answer the question actually asked.

- 1 Add the ratio parts: $2 + 3 + 7 = 12$.

why: The total is divided into this many equal parts.

- 2 One part: $£600 \div 12 = £50$.

why: Divide the total by the number of parts.

- 3 The difference between C and A is $7 - 2 = 5$ parts.

why: Work with the **difference** in parts rather than computing both shares.

- 4 So the difference is $5 \times £50 = £250$.

why: Check: A gets £100, C gets £350 ✓.

Where it goes wrong

- Giving C 's share (£350) instead of the difference - an offered option.
- Dividing by 3 (the number of people) instead of 12 (the number of parts).

Takeaway Find the value of one part first; then read carefully whether a share, a difference or a total is wanted.

Question 7 SEQ-02 · Sequences and series · spec MM2.3

Find the sum to infinity of the geometric series $12 - 4 + \frac{4}{3} - \frac{4}{9} + \dots$

Answer: C

$$a = 12, r = -\frac{1}{3}. S_{\infty} = \frac{12}{1 + \frac{1}{3}} = \frac{12}{4/3} = 9.$$

Full method - how and why

A sum to infinity. Find the ratio (including its sign) and check $|r| < 1$ before using the formula.

- Find r by dividing consecutive terms: $\frac{-4}{12} = -\frac{1}{3}$.
why: Always divide a term by the one before it; the alternating signs mean r is negative.
- Check convergence: $|\frac{1}{3}| < 1$ ✓.
why: The sum to infinity only exists when $|r| < 1$; otherwise the formula is meaningless.
- Apply $S_{\infty} = \frac{a}{1-r} = \frac{12}{1 - (-\frac{1}{3})} = \frac{12}{1 + \frac{1}{3}}$.
why: $1 - r$ with a negative r becomes $1 + \frac{1}{3}$ - the double negative is the main trap.
- Divide: $12 \times \frac{3}{4} = 9$.
why: Dividing by a fraction is multiplying by its reciprocal.

Where it goes wrong

- Using $r = +\frac{1}{3}$ and getting 18.
- Computing $\frac{12}{\frac{4}{3}}$ as 16 by multiplying instead of dividing.

Takeaway $S_{\infty} = \frac{a}{1-r}$; if r is negative the denominator grows, so the sum is smaller than $a/(1 - |r|)$.

Question 8 INT-02 · Integration · spec MM7.1

Find the area enclosed between the curve $y = x^2 - 4x$ and the x -axis.

Answer: F

Roots 0 and 4; the curve is below the axis between them. $\int_0^4 (x^2 - 4x) dx = \frac{64}{3} - 32 = -\frac{32}{3}$, so the area is $\frac{32}{3}$.

Full method - how and why

Area between a curve and the axis. The integral is negative here because the region is below the axis - the area is its magnitude.

- Find the limits: $x^2 - 4x = x(x - 4) = 0$ at $x = 0$ and $x = 4$.
why: The enclosed region runs between consecutive roots.
- Note the curve is **below** the axis there: at $x = 2$, $y = -4$.
why: Checking one interior value tells you the sign of the integral in advance.
- Integrate: $\int_0^4 (x^2 - 4x) dx = \left[\frac{x^3}{3} - 2x^2 \right]_0^4 = \frac{64}{3} - 32 = -\frac{32}{3}$.
why: $32 = \frac{96}{3}$, so $\frac{64-96}{3} = -\frac{32}{3}$.
- Take the magnitude: the area is $\frac{32}{3}$.
why: Areas are positive; the negative sign only records that the region lies below the axis.

Where it goes wrong

- Giving $-\frac{32}{3}$ as the area - it is offered precisely to catch this.
- Using limits 0 and 2 (the vertex) instead of the two roots.

Takeaway Integrate between the roots, then take the absolute value: a definite integral is a signed area.

Question 9 LOG-12 · Exponentials and logarithms · spec MM5.1

The curve $y = a^x$ passes through (2, 12) and (4, 108). Find a .

Answer: B

$$\frac{a^4}{a^2} = a^2 = \frac{108}{12} = 9, \text{ so } a = 3 \text{ (as } a > 0\text{)}.$$

Full method - how and why

Two points on an exponential curve. Dividing eliminates any constant and isolates a power of a .

- 1 Write both facts: $a^2 = 12$ and $a^4 = 108$.

why: Substitute each point into $y = a^x$.

- 2 Divide: $\frac{a^4}{a^2} = a^2 = \frac{108}{12} = 9$.

why: Dividing subtracts indices, giving a single power of a - the standard move with two points.

- 3 Solve: $a = 3$ (bases of exponentials are positive).

why: Reject $a = -3$: $y = a^x$ requires $a > 0$.

Where it goes wrong

- Answering 9 - that is a^2 , not a .
- Subtracting the y -values instead of dividing.

Takeaway Divide two points on an exponential curve; the index difference gives a clean power of the base.

Question 10 INT-03 · Integration · spec MM7.3

Given that $\int_1^5 f(x) dx = 7$ and $\int_3^5 f(x) dx = 2$, find $\int_1^3 (2f(x) - 1) dx$.

Answer: C

$$\int_1^3 f = 7 - 2 = 5. \text{ So } 2 \times 5 - (3 - 1) = 8.$$

Full method - how and why

Combining integrals over adjacent intervals, plus the linearity rules. No function is ever needed.

- 1 Use $\int_1^5 = \int_1^3 + \int_3^5$: so $\int_1^3 f = 7 - 2 = 5$.

why: Integrals over adjacent ranges add; this is the only way to get at the interval you need.

- 2 Split the required integral: $\int_1^3 (2f(x) - 1) dx = 2\int_1^3 f(x) dx - \int_1^3 1 dx$.

why: Constants come out; sums split - both are properties of the integral.

- 3 Evaluate the second piece: $\int_1^3 1 dx = 3 - 1 = 2$.

why: Integrating 1 over an interval gives its length.

- 4 Combine: $2(5) - 2 = 8$.

why: Careful with the order of operations.

Where it goes wrong

- Forgetting the -1 term and answering 10.
- Using $\int_1^3 f = 7 + 2 = 9$ instead of subtracting.

Takeaway $\int_a^c = \int_a^b + \int_b^c$, constants factor out, and $\int_a^b 1 dx = b - a$.

Question 11 SEQ-06 · Sequences and series · spec MM2.2

The sum of the first n terms of a series is $S_n = 2n^2 + 3n$. Find the 10th term.

Answer: F

$$u_{10} = S_{10} - S_9 = (200 + 30) - (162 + 27) = 230 - 189 = 41.$$

Full method - how and why

Given a formula for S_n , individual terms come from the difference of consecutive sums.

1 Use $u_{10} = S_{10} - S_9$.

why: The 10th term is what the total gains when you add it - this works for any series, arithmetic or not.

2 $S_{10} = 2(100) + 30 = 230$.

why: Substitute $n = 10$.

3 $S_9 = 2(81) + 27 = 189$.

why: Substitute $n = 9$; $2 \times 81 = 162$.

4 Subtract: $230 - 189 = 41$.

why: Both wrong intermediate values (230 and 189) are offered as distractors.

Where it goes wrong

- Answering 230 (that is the sum of ten terms, not the tenth term).
- Substituting $n = 10$ into a 'term' formula that was never given.

Takeaway $u_n = S_n - S_{n-1}$ recovers terms from a sum formula.

Question 12 TRG-04 · Trigonometric functions and equations · spec MM4.6

Find the sum of all solutions of $\tan x = \sqrt{3}$ in the interval $0 \leq x \leq 2\pi$.

Answer: D

$$x = \frac{\pi}{3} \text{ and } x = \frac{\pi}{3} + \pi = \frac{4\pi}{3}. \text{ Sum } \frac{5\pi}{3}.$$

Full method - how and why

$\tan$ has period π , so solutions come one per period. Find the first, add π , then check the interval.

1 The principal solution: $\tan x = \sqrt{3}$ at $x = \frac{\pi}{3}$.

why: From the exact values: $\tan 60^\circ = \sqrt{3}$.

2 $\tan$ repeats every π , so the next solution is $\frac{\pi}{3} + \pi = \frac{4\pi}{3}$.

why: Unlike $\sin$ and $\cos$, $\tan$ has period π , not 2π .

3 Check the interval $[0, 2\pi]$: $\frac{\pi}{3}$ and $\frac{4\pi}{3}$ are in it; the next, $\frac{7\pi}{3}$, is not.

why: $\frac{7\pi}{3} > 2\pi$, so stop there.

4 Sum: $\frac{\pi}{3} + \frac{4\pi}{3} = \frac{5\pi}{3}$.

why: Common denominator 3.

Where it goes wrong

- Using period 2π and finding only $\frac{\pi}{3}$.
- Including $\frac{7\pi}{3}$ and answering $\frac{7\pi}{3}$ or more - which is offered.

Takeaway $\tan$ repeats every π : solutions are $x_0 + k\pi$, so there are usually two in a 2π interval.

Question 13 DIF-07 · Differentiation · spec MM6.3

The curve $y = ax^2 + bx$ has a stationary point at $(2, -8)$. Find $a + b$.

Answer: A

$$y' = 2ax + b = 0 \text{ at } x = 2: b = -4a. y(2) = 4a + 2b = 4a - 8a = -4a = -8 \Rightarrow a = 2, b = -8. a + b = -6.$$

Full method - how and why

A stationary point gives two conditions: the derivative vanishes there, and the point is on the curve.

- 1 Differentiate: $y' = 2ax + b$. At $x = 2$: $4a + b = 0$, so $b = -4a$.

why: Stationary means the gradient is zero at that x .

- 2 The point is on the curve: $y(2) = 4a + 2b = -8$.

why: Substitute $x = 2$ into the original function and set it equal to the given y .

- 3 Substitute $b = -4a$: $4a - 8a = -4a = -8$, so $a = 2$ and $b = -8$.

why: One equation in one unknown after substitution.

- 4 Compute $a + b = 2 + (-8) = -6$.

why: The question asks for the sum.

Where it goes wrong

- Using only one of the two conditions.
- Sign error giving $a = -2$ and the answer 6.

Takeaway 'Stationary point at (p, q) ' is two equations: $f'(p) = 0$ and $f(p) = q$.

Question 14 PRO-10 · Probability · spec M7.7

A fair die is thrown repeatedly until a six appears. Find the probability that the first six appears on the third throw.

Answer: A

$$\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} = \frac{25}{216}.$$

Full method - how and why

'First success on the k th trial' means failure, failure, ..., then success - a chain of independent events.

- 1 The first two throws must be non-sixes: probability $\frac{5}{6}$ each.

why: 'First six on the third' forbids a six on throws one and two.

- 2 The third throw must be a six: probability $\frac{1}{6}$.

why: Exactly one way to succeed on that throw.

- 3 Multiply: $\left(\frac{5}{6}\right)^2 \times \frac{1}{6} = \frac{25}{216}$.

why: Independent throws, so probabilities multiply.

Where it goes wrong

- Answering $\frac{1}{6}$ by ignoring the first two throws.
- Using $\left(\frac{1}{6}\right)^3 = \frac{1}{216}$, which is the probability of **three** sixes.

Takeaway First success on trial k : $(1 - p)^{k-1}p$ - failures first, then the one success.

Question 15 LOG-17 · Exponentials and logarithms · spec MM5.3

The equation $9^x - k \cdot 3^x + 9 = 0$ has exactly one real solution. Find the value of k .

Answer: E

With $y = 3^x > 0$: $y^2 - ky + 9 = 0$. Exactly one solution for x means exactly one positive root: a repeated root ($k^2 = 36$, product $9 > 0$ so the roots have the sign of k): $k = 6$ (repeated root $y = 3$, $x = 1$). $k = -6$ gives $y = -3$, no solutions. (Two distinct positive roots would give two solutions.)

Full method - how and why

A quadratic in 3^x with a parameter. 'Exactly one real solution' is a condition on the **positive** roots, not simply on the discriminant.

- 1 Substitute $y = 3^x$, noting $y > 0$: $y^2 - ky + 9 = 0$.

why: $9^x = (3^x)^2$. The restriction $y > 0$ is the heart of the question.

- 2 The product of the roots is $9 > 0$, so the roots have the same sign; their sum is k , so both are positive exactly when $k > 0$.

why: Sum and product of roots decide the signs without solving.

- 3 For exactly one value of x , we need exactly one positive root, which (given both roots share a sign) means a **repeated** root: discriminant $k^2 - 36 = 0$, so $k = \pm 6$.

why: Two distinct positive roots would give two solutions for x ; two negative roots would give none.

- 4 Check the signs: $k = 6$ gives the repeated root $y = 3 > 0$, so $x = 1$ - exactly one solution. $k = -6$ gives $y = -3 < 0$, so no solutions at all. Hence $k = 6$.

why: The final sign check is what rules out -6 , which is offered.

Where it goes wrong

- Answering ± 6 from the discriminant alone.
- Forgetting that y must be positive and counting $y = -3$ as a solution.

Takeaway With $y = a^x$, translate conditions on x into conditions on the **positive** roots in y , using sum and product to check signs.

Question 16 QUA-14 · Quadratic functions · spec MM1.3

The equation $x^2 - 2mx + m + 6 = 0$ has two distinct positive roots. Find the complete set of values of m .

Answer: B

Distinct real roots: $4m^2 - 4(m + 6) > 0 \Rightarrow m^2 - m - 6 > 0 \Rightarrow (m - 3)(m + 2) > 0$, so $m < -2$ or $m > 3$. Both roots positive needs sum $2m > 0$ and product $m + 6 > 0$, so $m > 0$. Combined: $m > 3$.

Full method - how and why

'Two distinct **positive** roots' is three conditions at once: discriminant positive, sum of roots positive, product of roots positive. All three must hold.

- 1 Discriminant: $4m^2 - 4(m + 6) > 0 \Rightarrow m^2 - m - 6 > 0 \Rightarrow (m - 3)(m + 2) > 0$, so $m < -2$ or $m > 3$.

why: Distinctness and reality come from $b^2 - 4ac > 0$; this alone allows negative m .

- 2 Sum of roots $= 2m > 0$, so $m > 0$.

why: If both roots are positive their sum is positive. This is the condition that kills the $m < -2$ branch.

- 3 Product of roots $= m + 6 > 0$, so $m > -6$.

why: If both roots are positive their product is positive; this also rules out one positive and one negative root.

- 4 Intersect all three: $(m < -2 \text{ or } m > 3)$ and $m > 0$ and $m > -6$ gives $m > 3$.

why: Take the overlap on a number line - the answer must satisfy every condition simultaneously.

Where it goes wrong

- Stopping after the discriminant and choosing $m < -2$ or $m > 3$ - which is offered.
- Forgetting that sum **and** product must both be positive: either one alone is not enough.

Takeaway Both roots positive $\Leftrightarrow$ discriminant ≥ 0 , sum > 0 and product > 0 ; sign conditions on roots always come in this package.

Question 17 GRA-13 · Graphs of functions · spec MM8.7

For how many values of k does the equation $|x^2 - 4| = k$ have exactly three real solutions?

Answer: A

The graph of $y = |x^2 - 4|$ is a W-shape with minima 0 at $x = \pm 2$ and a local maximum 4 at $x = 0$. A horizontal line gives three solutions only when it passes through the local maximum: $k = 4$. One value.

Full method - how and why

Sliding a horizontal line across a modulus graph. Three solutions happen only at the height of the local maximum.

- 1 Sketch $y = |x^2 - 4|$: the parabola $x^2 - 4$ with the part between $x = -2$ and $x = 2$ flipped above the axis.

why: Reflecting the negative part gives a W-shape with two zeros and a hump.

- 2 Identify the key heights: zeros at $x = \pm 2$, and a local maximum of 4 at $x = 0$.

why: At $x = 0$, $|0 - 4| = 4$ - the top of the central hump.

- 3 Count intersections with $y = k$: for $0 < k < 4$ there are four; for $k > 4$ there are two; for $k = 0$ there are two; for $k = 4$ the line passes through the hump's peak **and** cuts the outer arms: three.

why: At the peak, two of the four intersections merge into one.

- 4 So exactly one value of k (namely 4) gives three solutions.

why: The question asks how many values of k , not how many solutions.

Where it goes wrong

- Answering 4 (the value of k) instead of 1 (the number of such k).
- Forgetting $k = 0$ gives two solutions, not three.

Takeaway Slide the horizontal line and note the heights where the count changes: those are the turning values.

Question 18 GRA-14 · Graphs of functions · spec MM8.1

The graph of $y = \sqrt{x}$ is transformed to $y = \sqrt{4-x}$. Which sequence of transformations achieves this?

Answer: E

$\sqrt{4-x} = \sqrt{-(x-4)}$: reflect $y = \sqrt{x}$ in the y -axis to get $\sqrt{-x}$, then translate 4 to the right. (Translating first then reflecting gives $\sqrt{-x-4}$.)

Full method - how and why

Order matters when combining a reflection with a translation. Rewrite the target to reveal the correct sequence.

- 1 Rewrite the target: $\sqrt{4-x} = \sqrt{-(x-4)}$.
why: Factoring out the -1 exposes both operations: a negation of x and a shift.
- 2 Reading it as $f(-(x-4))$ where $f(x) = \sqrt{x}$: first reflect in the y -axis (giving $\sqrt{-x}$), then translate 4 right (replacing x by $x-4$).
why: For nested transformations, the operation applied directly to x (here the shift) is done **last**.
- 3 Check with a point: $y = \sqrt{4-x}$ passes through $(4, 0)$; $\sqrt{-x}$ passes through $(0, 0)$, which moves to $(4, 0)$ after shifting right ✓.
why: A point check settles order questions decisively.
- 4 Verify the other order fails: translating first gives $\sqrt{x-4}$, then reflecting gives $\sqrt{-x-4}$ - not the target.
why: Different sequences give genuinely different graphs.

Where it goes wrong

- Choosing 'translate then reflect', which gives $\sqrt{-x-4}$.
- Reflecting in the x -axis instead of the y -axis.

Takeaway Rewrite as $f(\pm(x-h))$ and test a point; with mixed reflections and shifts the order is not interchangeable.

Question 19 RAT-09 · Ratio, proportion and percentages · spec M3.9

p is directly proportional to $\sqrt{q}$, and q is inversely proportional to r . Which of the following is true?

Answer: D

$$p = a\sqrt{q} \text{ and } q = \frac{b}{r}, \text{ so } p = a\sqrt{\frac{b}{r}} = \frac{a\sqrt{b}}{\sqrt{r}}.$$

Full method - how and why

Chaining two proportionality statements. Write each with a constant and substitute.

- 1 Write them algebraically: $p = a\sqrt{q}$ and $q = \frac{b}{r}$ for constants $a, b > 0$.
why: 'Directly proportional' means a constant multiple; 'inversely' means a constant over.
- 2 Substitute: $p = a\sqrt{\frac{b}{r}} = \frac{a\sqrt{b}}{\sqrt{r}}$.
why: The square root distributes over the quotient.
- 3 Since $a\sqrt{b}$ is a constant, $p = \frac{\text{constant}}{\sqrt{r}}$: p is inversely proportional to $\sqrt{r}$.
why: Collect all the constants into one; only the r -dependence matters.

Where it goes wrong

- Forgetting the square root and concluding $p \propto \frac{1}{r}$ - an offered option.
- Losing the 'inversely' and choosing $p \propto \sqrt{r}$.

Takeaway Chain proportionalities by substituting; the accumulated constants never affect the final relationship.

Question 20 LIN-08 · Straight lines · spec MM3.1

For which value of k are the lines $kx + 3y = 2$ and $2x - (k - 1)y = 5$ perpendicular?

Answer: A

Gradients $-\frac{k}{3}$ and $\frac{2}{k-1}$. Perpendicular when the product is -1 : $\frac{-2k}{3(k-1)} = -1 \Rightarrow 2k = 3k - 3 \Rightarrow k = 3$.

Full method - how and why

A perpendicularity condition with the parameter in both equations. Write both gradients, multiply, set to -1 .

- 1 First line $kx + 3y = 2$ has gradient $-\frac{k}{3}$.
why: Using $-\frac{a}{b}$ avoids rearranging.
- 2 Second line $2x - (k - 1)y = 5$ has gradient $-\frac{2}{-(k-1)} = \frac{2}{k-1}$.
why: Here $b = -(k - 1)$, so the two minus signs cancel.
- 3 Set the product to -1 : $-\frac{k}{3} \times \frac{2}{k-1} = -1$, i.e. $\frac{-2k}{3(k-1)} = -1$.
why: Perpendicular lines have gradients multiplying to -1 .
- 4 Solve: $-2k = -3(k - 1) \Rightarrow -2k = -3k + 3 \Rightarrow k = 3$.
why: Check: gradients -1 and 1 , product -1 ✓.

Where it goes wrong

- Mishandling the sign of b in the second line and getting $k = -3$ - which is offered.
- Setting the product to $+1$ or the gradients equal (that would be the parallel condition).

Takeaway Read gradients as $-\frac{a}{b}$ straight from $ax + by = c$, watching the sign of b carefully.